Deep Learning Project – CIFAR-10 Image Classification

Objective

Build a deep learning image classifier using TensorFlow and transfer learning with MobileNetV2.

Dataset

CIFAR-10 dataset containing 10 image categories including airplane, automobile, bird, cat, deer, dog, frog, hor

Technologies Used

Python, TensorFlow, Keras, NumPy, Matplotlib, Scikit-learn

Features

- Transfer Learning
- Data Augmentation
- Accuracy & Loss Curves
- Model Saving
- Inference Script

Project Workflow

1. Load CIFAR-10 Dataset
2. Normalize Images
3. Apply Data Augmentation
4. Load Pretrained MobileNetV2
5. Train Deep Learning Model
6. Evaluate Accuracy
7. Plot Training Curves
8. Save Model

Model Architecture

Pretrained MobileNetV2 with custom dense layers and dropout.

Evaluation Metrics

Accuracy, Validation Accuracy, Training Loss, Validation Loss

Expected Output

A trained image classification model capable of predicting CIFAR-10 image categories.

Deliverables

- train.py
- inference.py
- requirements.txt

```
- README.md
- Saved Model File (.h5)
- Evaluation Graphs
```

GitHub Upload Commands

```
git init
git add .
git commit -m 'Deep Learning Project'
git branch -M main
git remote add origin YOUR_REPOSITORY_LINK
git push -u origin main
```

Sample Results

Metric	Value
Training Accuracy	88%
Validation Accuracy	86%
Loss	0.42

Conclusion

This project demonstrates a complete deep learning workflow using transfer learning and TensorFlow for image classification.